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CSA0911

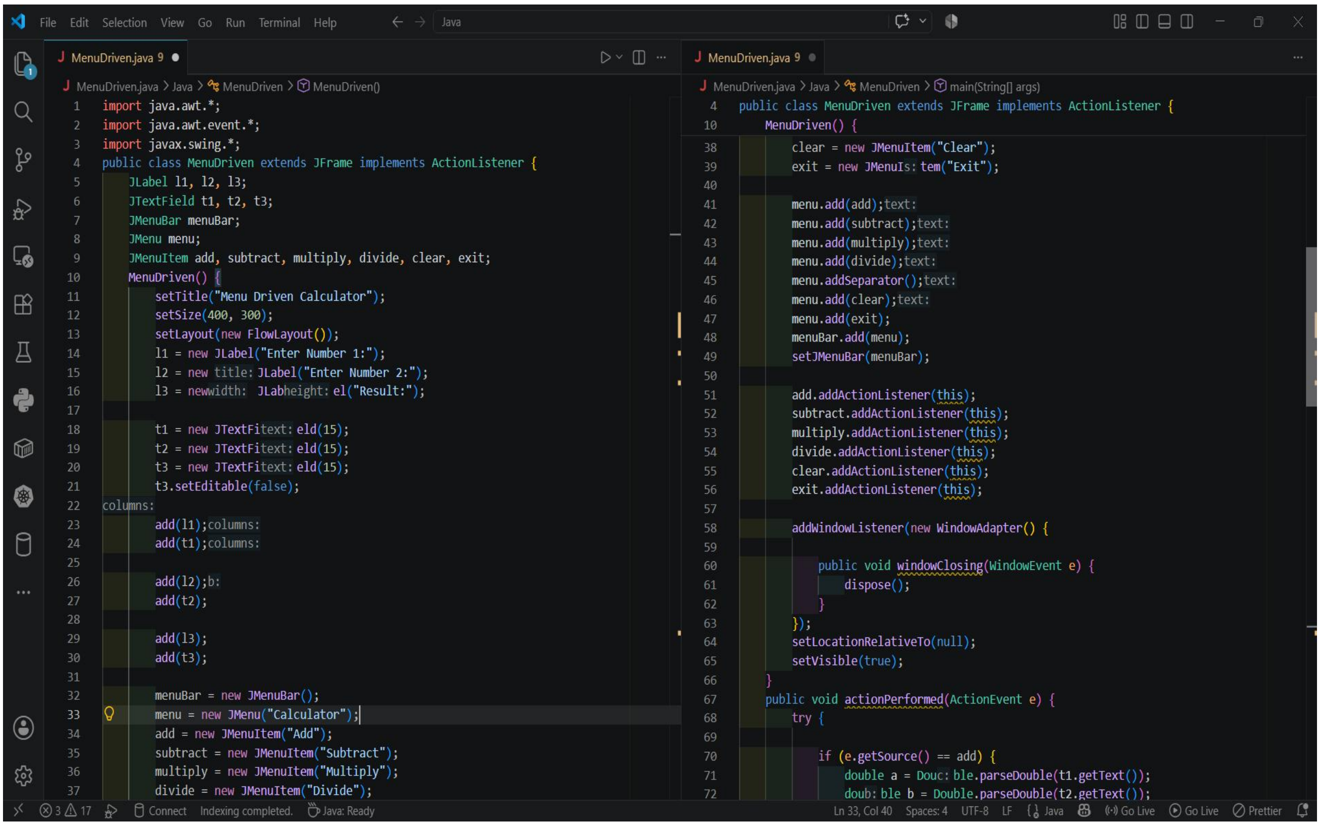
JAVA

CO4 AT2 Q2

Q2 menu driven 1 mark

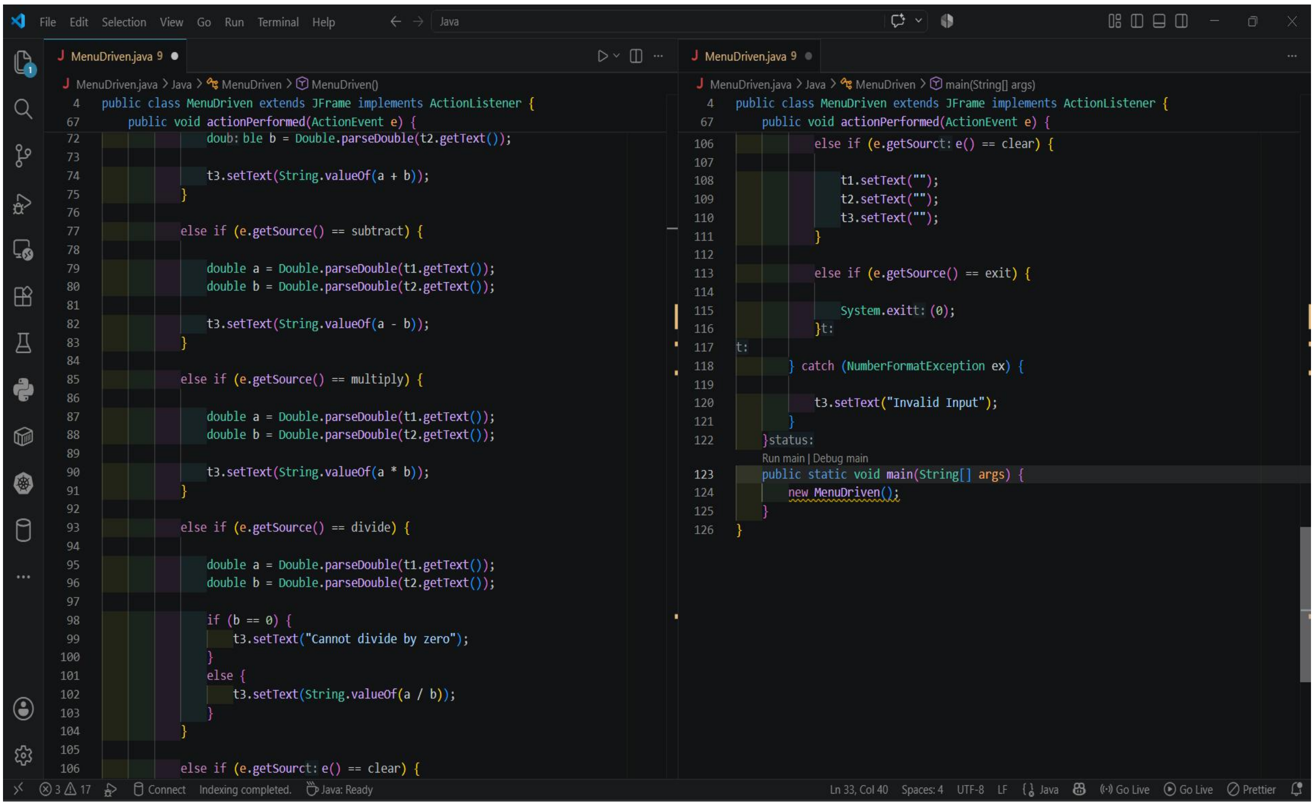
Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

CODE:



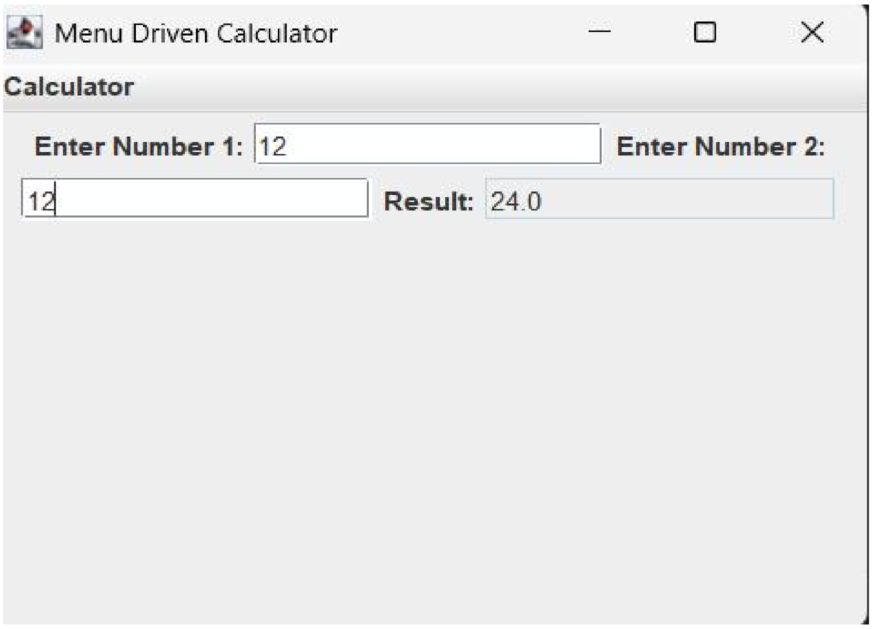
```java
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
public class MenuDriven extends JFrame implements ActionListener {
    JLabel l1, l2, l3;
    JTextField t1, t2, t3;
    JMenuBar menuBar;
    JMenu menu;
    JMenuItem add, subtract, multiply, divide, clear, exit;
    MenuDriven() {
        setTitle("Menu Driven Calculator");
        setSize(400, 300);
        setLayout(new FlowLayout());
        l1 = new JLabel("Enter Number 1:");
        l2 = new title: JLabel("Enter Number 2:");
        l3 = newwidth:  JLabheight: el("Result:");

        t1 = new JTextFitext: eld(15);
        t2 = new JTextFitext: eld(15);
        t3 = new JTextFitext: eld(15);
        t3.setEditable(false);
columns:
        add(l1);columns:
        add(t1);columns:

        add(l2);b:
        add(t2);

        add(l3);
        add(t3);

        menuBar = new JMenuBar();
        menu = new JMenu("Calculator");
        add = new JMenuItem("Add");
        subtract = new JMenuItem("Subtract");
        multiply = new JMenuItem("Multiply");
        divide = new JMenuItem("Divide");
```

```java
public class MenuDriven extends JFrame implements ActionListener {
    MenuDriven() {
        clear = new JMenuItem("Clear");
        exit = new JMenuIs: tem("Exit");

        menu.add(add);text:
        menu.add(subtract);text:
        menu.add(multiply);text:
        menu.add(divide);text:
        menu.addSeparator();text:
        menu.add(clear);text:
        menu.add(exit);
        menuBar.add(menu);
        setJMenuBar(menuBar);

        add.addActionListener(this);
        subtract.addActionListener(this);
        multiply.addActionListener(this);
        divide.addActionListener(this);
        clear.addActionListener(this);
        exit.addActionListener(this);

        addWindowListener(new WindowAdapter() {

            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });
        setLocationRelativeTo(null);
        setVisible(true);
    }
    public void actionPerformed(ActionEvent e) {
        try {

            if (e.getSource() == add) {
                double a = Douc: ble.parseDouble(t1.getText());
                doub: ble b = Double.parseDouble(t2.getText());
```

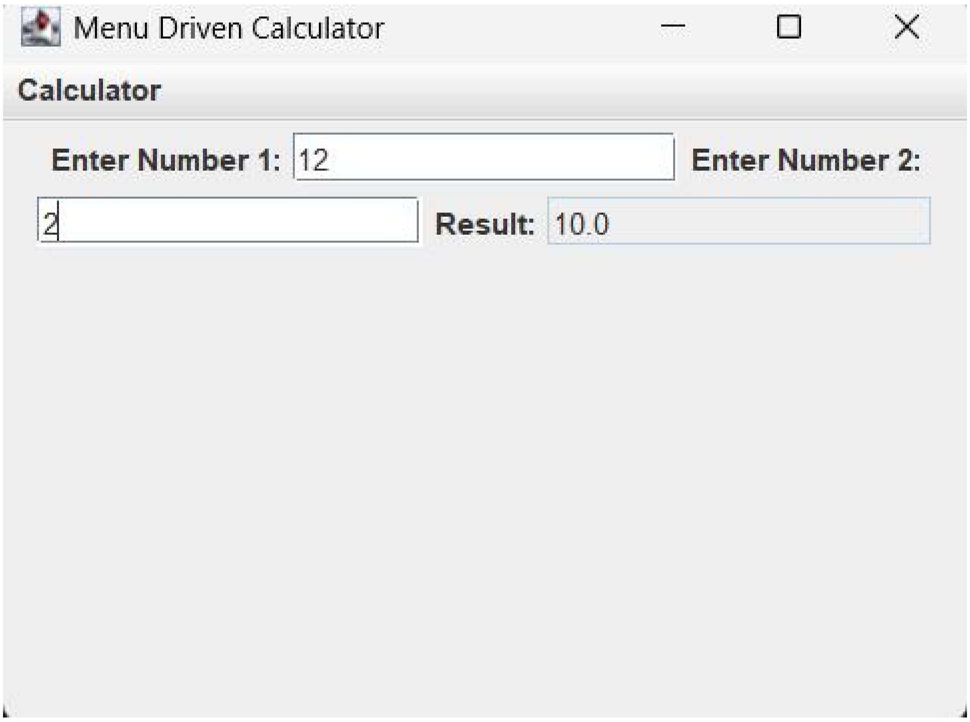


OUTPUT:

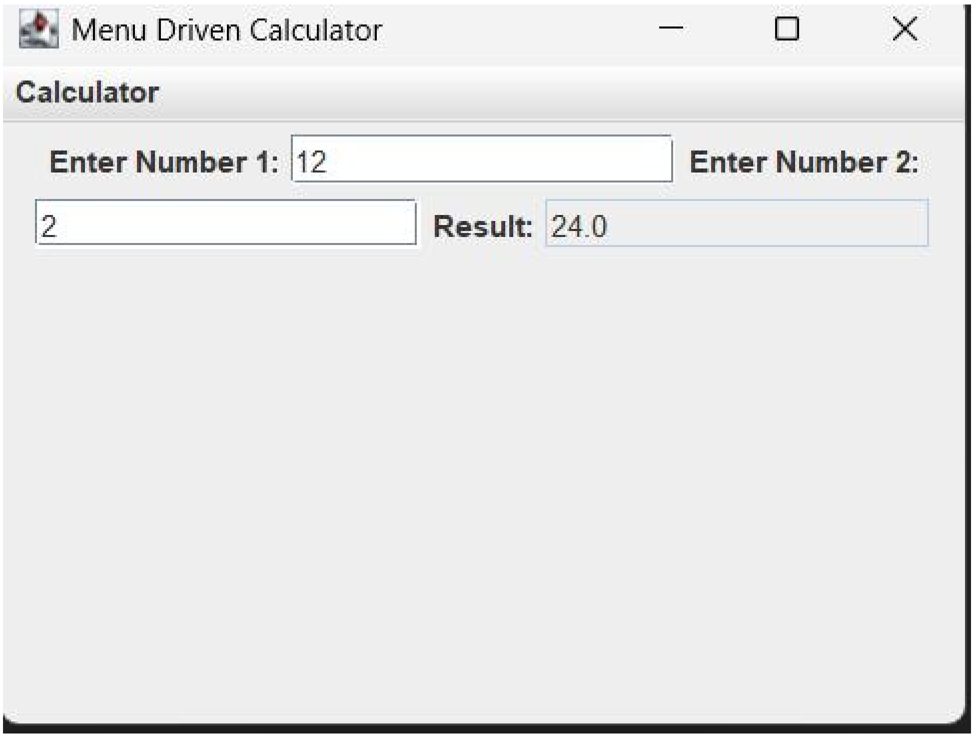
Addition:



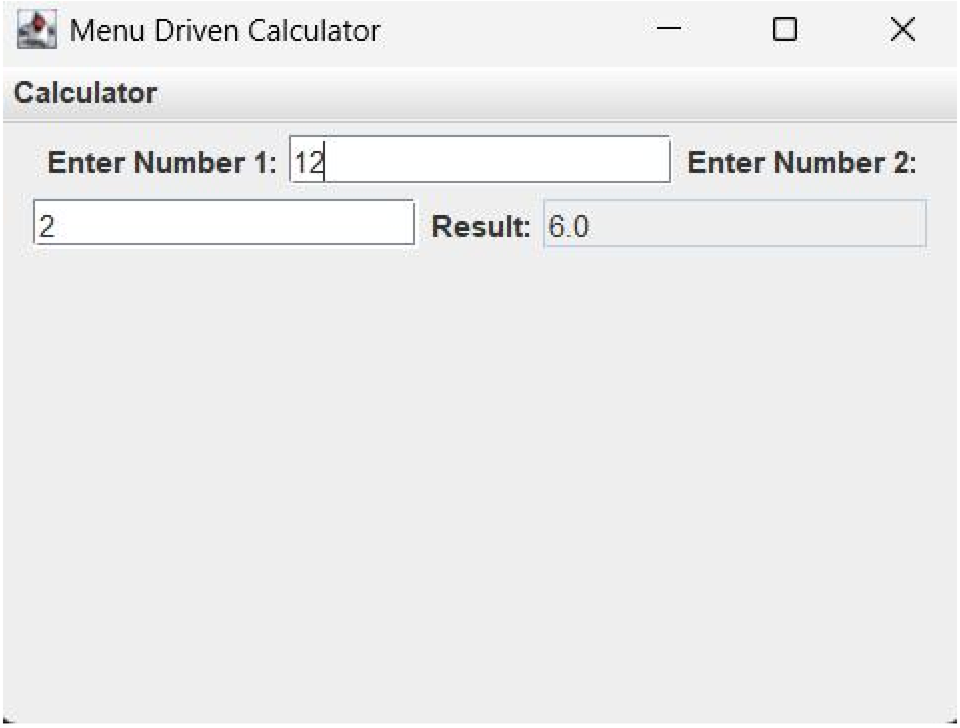
Subtraction:



Multiplication:



Division:


Menu Driven Calculator
Calculator
Enter Number 1:
12
Enter Number 2:
2
Result:
6.0